

E-Commerce Profitability Analysis | SQL| Python | Power BI

1. Business Problem

Despite strong sales growth, the business faced inconsistent profitability due to high discounts, low-margin products, and loss-making orders. There was limited visibility into how discounts, categories, and regions were impacting overall profit performance

2. Objective

Analyze the impact of discounts, product categories, and regional sales on profitability to identify key drivers of profit loss and revenue growth opportunities.

3. Tools & Project Links:

SQL : [Link](#) | Python: [Link](#) | Power BI: [Link](#)

4. Analysis:

- Performed EDA
- Conducted correlation analysis
- Performed discount threshold analysis
- Conducted discount vs profit margin analysis
- Performed category vs profit margin analysis
- Conducted discount bucket analysis
- Performed category-wise profitability analysis
- Conducted time-based sales and profit analysis

5. Key Insights

- Low discount range (0–10%) generated the highest profit margin at **30.05%**, indicating stronger profitability with minimal discounting
- Discounts above 40% resulted in negative profitability, with profit margin dropping to **-27.01%**, highlighting a high-risk discount zone
- **13.37% of total orders** were in the high-risk discount zone (>40%), contributing significantly to profit loss
- Body Care was the best-performing category, generating total profit of **\$590,907.78**
- Home and Accessories was the most loss-making category, recording total profit of **-\$57,187.64**

6. Recommendations

- Keep discounts below the identified high-risk threshold to maintain healthy profit margins
- Reduce aggressive discounting in low-margin product categories
- Focus promotional campaigns on high-profit product groups
- Monitor high-discount transactions regularly to control profitability loss
- Implement targeted discount strategies instead of blanket discounting across all products